

ZADUI, China

WMO: 560180

Lat: 32.8934N

Lon: 95.2949E

Elev: 4068

StdP: 61.10

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-18.9	-32.0	0.3	-8.7	-30.2	0.4	-8.1	10.4	4.3	9.3	1.6	1.0	180	0.701

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	20.6	9.5	19.1	9.2	17.8	8.7	11.0	17.8	10.4	16.9	9.9	15.9	2.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.5	11.5	13.2	8.0	11.1	12.9	7.5	10.8	12.6	45.3	18.0	43.4	17.0	41.7	16.1	14.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.5	5.3	DB	-24.8	23.4	2.5	1.4	-26.7	24.4	-28.1	25.2	-29.5	26.0	-31.4	27.0	(4)
(5)			WB	-25.5	12.3	2.3	0.8	-27.1	12.9	-28.5	13.3	-29.8	13.7	-31.5	14.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-9.1	-6.5	-2.6	1.9	6.2	10.1	12.3	11.8	9.0	2.7	-3.2	-7.6	(6)	
	DBStd	8.00	3.80	3.48	2.93	2.58	2.42	2.48	1.92	2.02	2.38	3.06	2.61	3.50	(7)	
	HDD10.0	3063	592	462	390	242	120	29	4	7	46	227	397	547	(8)	
	HDD18.3	5917	851	696	648	492	375	248	188	202	281	486	647	805	(9)	
	CDD10.0	188	0	0	0	0	3	31	75	63	16	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	3	0	0	0	0	0	1	1	1	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	1.9	1.9	2.1	2.3	2.1	2.0	1.7	1.7	1.7	1.6	1.7	1.9	1.8	(14)	
(15) Precipitation	PrecAvg	531	6	6	9	15	43	116	115	98	83	30	5	4	(15)	
	PrecMax	694	26	13	26	36	94	196	216	157	165	129	19	22	(16)	
	PrecMin	412	0	0	0	2	8	48	47	40	39	0	0	0	(17)	
	PrecStd	73	6	4	6	9	18	29	39	27	31	25	4	5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	7.0	11.5	15.5	20.2	21.0	22.5	22.8	19.6	16.4	8.7	6.6	(19)	
		MCWB	-3.1	-2.9	-0.6	1.7	5.8	9.2	10.6	9.7	8.9	4.8	-1.5	-3.1	(20)	
	2%	DB	3.6	4.7	8.2	12.6	16.2	18.5	20.3	20.7	17.7	12.6	6.4	4.1	(21)	
		MCWB	-4.8	-4.0	-2.0	0.9	4.5	8.7	10.0	9.5	8.3	3.7	-2.5	-4.5	(22)	
	5%	DB	0.7	2.9	6.1	10.5	14.1	16.9	19.0	19.1	16.0	10.4	4.9	1.8	(23)	
		MCWB	-6.4	-4.9	-2.7	0.2	3.9	8.2	9.4	9.3	7.6	2.4	-3.2	-5.6	(24)	
	10%	DB	-1.7	1.1	4.3	8.7	12.3	15.4	17.6	17.4	14.4	8.5	3.4	-0.2	(25)	
		MCWB	-7.4	-5.8	-3.4	-0.5	3.2	7.8	9.1	8.9	7.1	1.7	-3.9	-6.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-2.6	-0.2	3.1	7.8	11.0	12.0	11.8	10.2	6.5	-0.7	-2.6	(27)	
		MCDB	6.3	5.5	8.5	10.5	14.2	17.4	19.5	18.7	17.1	12.1	5.4	5.9	(28)	
	2%	WB	-4.6	-3.7	-1.3	2.0	6.3	10.1	11.2	10.8	9.3	5.1	-1.7	-4.3	(29)	
		MCDB	2.7	3.6	6.1	9.4	12.8	16.3	18.1	17.5	15.8	10.5	4.6	3.5	(30)	
	5%	WB	-6.0	-4.6	-2.1	1.2	5.4	9.3	10.6	10.2	8.6	3.9	-2.6	-5.4	(31)	
		MCDB	-0.1	1.9	4.5	8.3	11.3	15.0	17.1	16.7	14.5	8.5	3.6	1.3	(32)	
	10%	WB	-7.1	-5.5	-2.9	0.4	4.5	8.6	10.1	9.7	7.9	2.8	-3.4	-6.5	(33)	
		MCDB	-2.1	0.3	3.2	6.9	10.0	13.7	15.9	15.7	13.1	6.8	2.5	-0.7	(34)	
(35) Mean Daily Temperature Range	MDBR		13.7	13.2	12.6	12.0	10.9	9.8	10.6	11.0	10.9	11.0	13.4	14.5	(35)	
	5% DB	MCDBR	17.7	16.9	16.0	15.8	15.0	12.4	13.4	14.4	13.4	13.8	15.6	17.8	(36)	
		MCWBR	11.2	10.2	8.4	6.9	5.9	4.6	4.9	5.2	5.4	6.0	8.5	11.0	(37)	
	5% WB	MCDBR	15.9	15.3	13.5	12.6	11.1	10.1	10.5	10.8	11.2	11.1	13.9	16.9	(38)	
		MCWBR	10.4	9.5	7.4	5.7	5.0	4.4	4.2	4.3	4.8	5.2	7.7	10.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.192	0.215	0.264	0.305	0.312	0.296	0.286	0.281	0.261	0.236	0.201	0.182	(40)	
	taud		2.597	2.523	2.383	2.294	2.362	2.521	2.607	2.610	2.636	2.627	2.639	2.678	(41)	
	Ebn at Noon		1073	1066	1020	982	972	982	990	996	1013	1026	1054	1076	(42)	
	Edh at Noon		75	90	114	131	124	106	96	94	88	81	71	65	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.37	4.23	5.35	6.25	6.42	5.85	5.89	5.42	4.86	4.27	3.91	3.33	(44)	
	RadStd		0.34	0.26	0.30	0.34	0.45	0.34	0.34	0.43	0.26	0.22	0.14	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.49	+1.19	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (4 neighbors)	+0.45	+0.73	-1.51	N/A	N/A	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page